Can AI See the Fear Behind the Anger?

— B3 as Intent Hypothesis Retention —

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Abstract

Current LLMs read context with high accuracy and generate appropriate responses. However, generating high-precision responses is not the same as constructing responses while retaining multiple utterance-purpose candidates. This paper treats this difference as a structural problem of interpretation.

This paper organizes the pathways of understanding others through the B-Series of COS, defining B1 (literal interpretation), B2 (contextual interpretation), and B3 (intent hypothesis retention). B3 is positioned as the process of constructing responses while retaining and updating multiple hypotheses about the speaker's intent, rather than converging on a single interpretation.

Within the author's range of observation, B3-like processing in this sense has not been sufficiently confirmed in current LLMs. This paper refers to this as the B3 Unobserved Hypothesis. The tendency to converge prematurely onto the most plausible interpretation is organized as single-interpretation lock, and the possibility that this is structurally related to candidate loss and the cognitive misalignment observed as D2 is examined.

This paper begins from N=1 observation. Drawing on the author's own observations, this paper organizes the B3 formation hypothesis and examines the possibility of external implementation using SCOPE. SCOPE can function as an inferential coordinate system for placing and continuing to observe multiple intent hypotheses. This paper examines the possibility that combining B3 and SCOPE could realize continued candidate retention as an external structure.

However, none of the claims presented in this paper have been verified. The existence, effectiveness, and feasibility of implementing B3 have not been confirmed. What this paper presents is not a verified theory, but a hypothesis awaiting verification. Its verification will require the construction of a habitat that includes an utterance-purpose cross-reference environment, a failure log collection environment, a long-term observation environment, an implementation environment, and a researcher network.

Keywords: B3, Intent Hypothesis Retention, Candidate Retention, Dialogue Systems, Human-AI Interaction, Cognitive Misalignment, SCOPE

0. Introduction

0-1. Is Your AI Really Understanding Your Users?

When someone says "I'm exhausted," what do you say back? Words of comfort? Permission to rest? Maybe they just want to be heard, or perhaps it's simply the opening of a conversation. We typically read the situation and pick one.

Current AI responds with impressive accuracy. "That sounds tough. Make sure to get some rest." And most of the time, that's enough. But this gap becomes visible in more complex situations. In a customer support context, for example, a sharp and angry tone may carry something other than anger. Panic, perhaps. A defensive reaction. Or a demand for someone to be held accountable. The possibilities are not singular.

The question is: how do we — humans and LLMs alike — handle that multiplicity? Even when the output looks the same, is the same thing happening internally? Could this blind spot be contributing to relationship friction and UX degradation?

This paper explores that question through the lens of cognitive architecture.

0-2. The Limitations of Current LLMs

Current LLMs read context from text input and return the most appropriate response. This capability has improved year by year, reaching near-human accuracy in many situations.

However, performing with high accuracy is not the same as retaining multiple interpretive possibilities.

Consider a situation where a single utterance supports multiple valid interpretations. "That's enough" — is it resignation? Anger? A report that the issue has been resolved? Current LLMs select the most probable one and respond accordingly. At least from observable interactions, a structure that retains multiple possibilities simultaneously while formulating a response is difficult to detect.

This "convergence to a single interpretation" causes no problems in most situations. However, in cases where user intent is complex, or where emotion and demand are intertwined, interpretive misalignment becomes more likely. That misalignment may appear small, but across repeated interactions it has the potential to accumulate as UX degradation and relational cost.

This paper treats this not as a problem of accuracy, but as a structural problem of interpretation.

0-3. The Hypothesis of This Paper

This paper focuses not on the accuracy of interpretation, but on its structure.

Current LLMs return responses with high accuracy. Whether their processing simultaneously retains multiple intent candidates, however, is a separate question. This paper organizes such interpretive structures through the B-Series of COS (Cognitive Operating System). Details are covered in Chapter 1.

B3, as defined in this paper, is not a function for correctly identifying intent. When someone says "Got it," are they genuinely understanding, reluctantly complying, or simply trying to end the conversation? B3 refers to the structure that retains these possibilities in parallel rather than collapsing them into one. This paper treats this function as B3 (Intent Hypothesis Retention) within the B-Series.

For now, this remains a hypothesis. Whether B3 actually exists, whether it can be implemented, and whether it is effective have not yet been verified. This paper is a record of observation and reasoning — a structural consideration of that question.

0-4. The Position of This Paper

This paper begins from N=1 observation. There is a single observer, and the current stage is limited to observation and reasoning.

The hypothesis is unverified. Whether B3 exists, whether it functions, whether it is effective — none of these have been confirmed.

Generalization is not yet possible. Additional observation and a verification environment are required, and those conditions are discussed in Chapter 8.

1. What Is the B-Series?

1-1. B1: Literal Interpretation

B1 is the layer that receives utterances at face value and generates responses accordingly. It follows the dictionary definitions of words, and context, emotion, or the speaker's state tend not to become targets of interpretation.

For "I'm tired," the utterance is processed as a statement of fact — that the speaker is in a state of fatigue — and a response is generated based on that result. For "Got it," it is processed as a report of understanding.

This processing is fast and stable. Interpretive variance is low, and consistent responses are possible. In B1, however, the utterance itself takes priority over the intent, emotion, or relational context behind it.

Such processing has the possibility of being observed in both humans and AI. Concrete examples will be addressed in later chapters.

1-2. B2: Contextual Interpretation

B2 is the layer that interprets utterances by referencing context and generates responses accordingly. It references the flow of conversation, emotional tone, and relational dynamics to determine what output to produce. In this processing, the basis for response generation is not the speaker's actual state, but rather how the utterance comes to be assigned meaning on the receiver's side.

For "I'm tired," if it is assigned the meaning that "a word of encouragement should be returned" based on the day's workload and the flow of conversation, a response is generated in that direction. For "Got it," the response varies depending on how meaning is assigned on the receiver's side. If it is assigned as sufficient understanding, the conversation moves forward. If assigned as insufficient understanding, a confirmation is made. If assigned as sarcasm, a response corresponding to that meaning is generated.

Whereas B1 processes the surface of an utterance, B2 determines its own output through meaning assignment on the receiver's side while referencing contextual information. This enables more natural and situationally appropriate responses.

Such processing has the possibility of being observed in both humans and AI. Concrete examples will be addressed in later chapters.

1-3. B3: Intent Hypothesis Retention

B3 is the layer that continues to hold multiple hypotheses about a speaker's utterance purpose, and generates responses based on them. The goal is not to correctly "guess" intent. What matters is not eliminating candidates.

The generation of multiple hypotheses itself is not a process unique to B3. This paper distinguishes between the generation of utterance-purpose candidates and the process of holding and updating them. The relationship with the inferential structure discussed later (SCOPE) will be examined again in Chapter 6. What B3 is responsible for is holding these candidates along the axis of utterance purpose, and constructing responses on that basis.

For an utterance such as "I understand," the following utterance-purpose candidates can be assumed: wanting to move forward because one has understood; wanting to end the conversation because further discussion seems pointless; wanting to avoid worsening the relationship; wanting time to think. Multiple possibilities like these can coexist.

These candidates are retained. New information may strongly support one hypothesis. However, the other hypotheses do not disappear immediately. No candidate is finalized, because a speaker's utterance purpose cannot be directly observed from the outside.

Whereas B2 is a process that determines one's own output based on meaning assigned from the receiver's side, B3 is a process that continues to hold the speaker's utterance purpose without eliminating it, and constructs responses while taking that plurality into account. These two processes differ fundamentally in direction.

What matters in B3 is not candidate generation but not eliminating candidates. As a result, when new information arrives later, it becomes possible that "it might have been that candidate after all." At the same time, the retained candidates may also influence response generation. A response that strongly presupposes only one candidate can make it difficult to continue the conversation if another candidate turns out to be correct. Constructing a response while holding multiple candidates may work to preserve room for the hypothesis to be updated later. The applied possibilities of this kind of response generation will be revisited in Chapter 7.

This paper does not provide a detailed explanation of SCOPE. The possibility that this kind of process occurs in humans or AI will be examined in a later chapter.

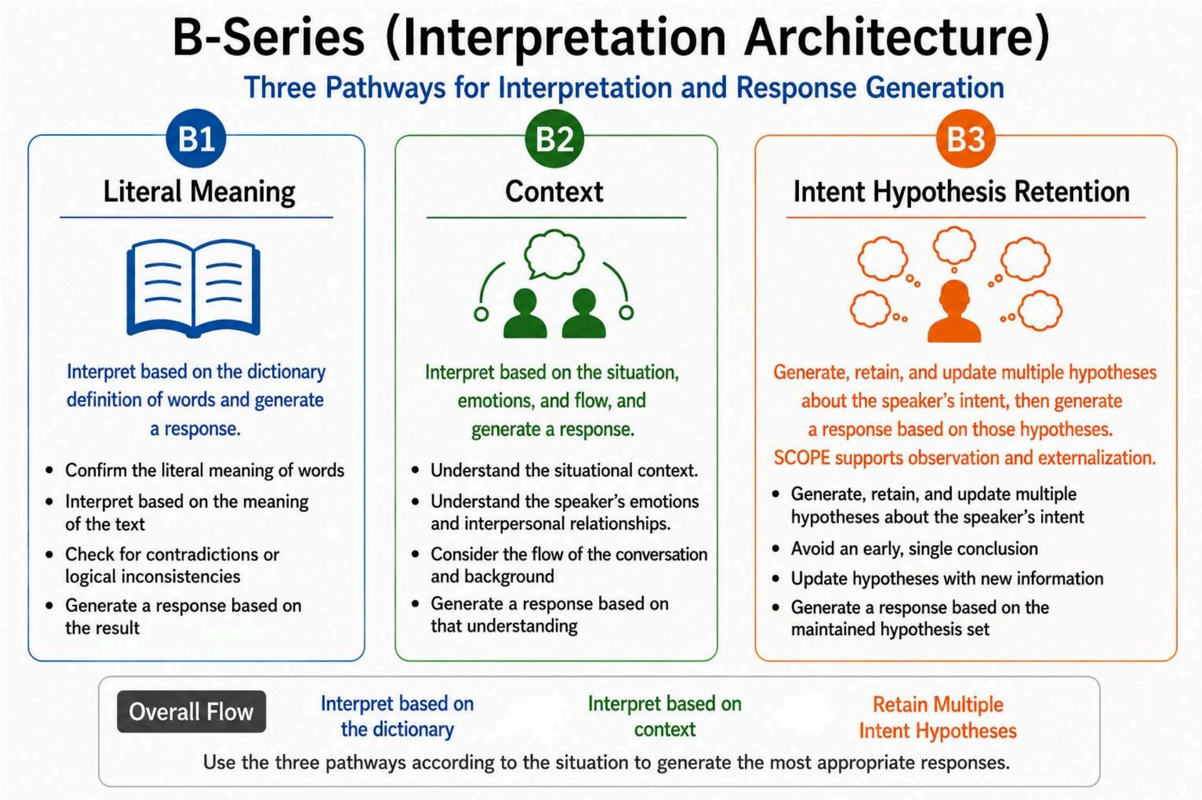
1-4. The Relationship Among the Three Layers

B1, B2, and B3 are not hierarchical upgrades that build upon one another in sequence. They coexist as layers with distinct roles.

B1 processes the surface of an utterance. B2 determines its own output through meaning assignment on the receiver's side while referencing contextual information. B3 continues to retain the speaker's communicative purpose without eliminating candidates, on the basis of SCOPE's observation. These three layers perform different processing on the same input.

B2 and B3 in particular differ in the direction of their processing. B2 is a process that constructs one's own output, while B3 is a process that retains the speaker's communicative purpose.

This paper also examines the possibility that differences exist in the tendencies with which each layer is utilized. Details will be addressed from Chapter 2 onward.



2. Why Do LLMs Appear to Stop at B2?

2-1. LLMs as B1-Origin Systems

LLMs begin processing from language input.

This is a structural characteristic, not a defect. Whereas humans receive information simultaneously through multiple channels such as vision, hearing, and bodily sensation, LLMs process input information as symbolically encoded data. Even in models capable of handling images and audio, that information is converted into representations within the model's internal space before being processed.

Regardless of input format, LLMs have a processing structure in which they do not directly access the internal states of their subjects, but instead make estimations based on externally expressed information.

In other words, LLMs have a structure in which intent is not "directly observed" but rather estimated from external information including text. This difference in starting point has the possibility of also influencing the subsequent interpretive process.

2-2. B2 Optimization

LLMs have dramatically improved their capacity for contextual inference through large-scale training.

Beyond B1-level processing that returns only the dictionary meanings of words, they possess the ability to integrate contextual information — such as the situation in which an utterance is placed, the flow of conversation, and the emotional tone of the exchange — to assign meaning to utterances accordingly and generate responses based on that meaning assignment. This is B2-level processing.

For example, for the input "I'm tired," B1-level processing returns a literal interpretation of being in a state of fatigue. In B2-level processing, a response is generated based on meaning assignment on the receiver's side, referencing the conversational context and the preceding exchange.

This capability has reached a level that is considered practically sufficient in current LLMs.

2-3. Single-Interpretation Lock

However, high-precision response candidate generation may be accompanied by a separate tendency. This is the tendency to converge early toward the most probable candidate among the multiple response candidates generated.

From a processing standpoint, this appears to be a natural tendency. Converging toward the most probable candidate, rather than continuing to hold multiple candidates, makes it easier to ensure response speed and consistency.

However, this convergence may produce one side effect. Response candidates other than the most probable one may become less likely to be referenced in subsequent processing. At the same time, the intent candidates underlying those response candidates may similarly become less likely to be referenced.

For example, if the system converges on "an empathetic response is appropriate" in response to "I'm tired," the candidate "a response that checks on the situation may be appropriate" may not carry over into the next turn.

This convergence tendency is referred to in this paper as single-interpretation lock. The problem of candidate attenuation will be examined in detail in 3-4.

2-4. The B3 Unobserved Hypothesis

As shown in the preceding sections, LLMs may generate multiple response candidates. However, whether the intent candidates underlying those response candidates are retained across turns and continuously updated is a separate question.

In this paper, this processing structure is defined as B3: the parallel and continuous retention of intent candidates. B3 refers not to the generation of candidates, but to a processing structure in which the purposes behind an utterance — what the other party is seeking, or what they are attempting to convey — are retained as multiple candidates that are not necessarily singular, and continuously updated on the basis of new information.

Within the author's observations, no sufficient evidence has been obtained that B3 processing in this sense exists in current LLMs. Each time a response is generated as output, the remaining intent candidates appear to become less likely to be referenced in subsequent processing.

However, this is not proof that B3 processing does not exist in LLMs. This is strictly a hypothesis based on the absence of confirmation within the author's observation range.

This hypothesis is referred to in this paper as the B3 Unobserved Hypothesis.

3. What Happens Without B3?

3-1. Interpreting "I'm Tired"

When presented with the utterance "I'm tired," current LLMs frequently generate responses offering sympathy or encouragement. An output such as "You must be exhausted. Please get some rest" appears to function adequately as B2 processing — reading the context and generating an appropriate response.

However, the question here is not whether the output is appropriate. The question is whether the intent candidates underlying the utterance "I'm tired" are carried forward into subsequent processing.

This utterance may carry multiple possible intent candidates: wanting rest, seeking empathy, wanting acknowledgment or praise, wanting support or something in return, wanting to move toward problem-solving, or simply wanting to express the feeling — these are only a portion of the candidates that can arise from a single utterance.

In B2 processing, a response is generated on the premise of the most probable intent candidate among these. Whether the remaining candidates are carried over into the next turn is a separate question. If the speaker then says "No, that's not what I meant," B2 processing produces a correction. However, that correction appears to be not an update of retained candidates, but a re-convergence toward a new single interpretation.

3-2. Interpreting "Why Did the Data Disappear?"

The following is a hypothetical scenario.

A speaker who noticed data loss during work inputs "Why did the data disappear?" In B2 processing, this utterance is likely interpreted as a question containing anger or frustration. A response such as "I apologize. Allow me to explain the steps" may be generated. This appears to function adequately as B2 processing.

However, multiple intent candidates may exist behind this utterance. Wanting to know the cause, wanting to restore the data, wanting to express anxiety or frustration, seeking compensation or relief, wanting someone to receive their emotions — these are only a portion of the candidates that can arise from a single utterance.

What is worth noting is that the surface-level anger or frustration alone may not be sufficient to explain the purpose of this utterance. There is a possibility that anxiety or urgency about the prospect of being unable to recover the data is present, and there is also a possibility that the speaker is seeking cause investigation or problem resolution. In B2 processing, a response may be generated as a reaction to the surface emotion. However, processing of the kind "they are angry, so calm them down" is not the retention of intent candidates — it is convergence toward the surface emotion.

As with the "I'm tired" example, the question is not whether the response is appropriate. The question is whether the intent candidates are carried forward into the next turn.

3-3. Conflating Anger with Intent

In the scenario presented in 3-2, suppose the LLM responds with "Please calm down. Allow me to explain the steps." This output appears at first glance to be an appropriate response that reads the speaker's state.

However, it is difficult to determine from this response alone that B3 processing has taken place. This response may have been generated as the result of output optimization. This is a typical example of B2 processing, and is a separate process from the retention of intent candidates.

Even if the resulting output appears the same, the underlying process may be fundamentally different.

In B3 processing, "anger" is treated as one piece of evidence for estimating intent. Anxiety about the possibility of being unable to recover the data, the desire to know the cause, the need for someone to receive their emotions — these candidates are retained while the response is constructed.

In B2 processing, the surface emotion serves as the primary cue for response generation, and the response is generated on that premise. Intent candidates may not be retained.

The question is not which output is "better." Whether candidates are retained affects processing in subsequent turns. If the speaker says "No, that's not what I meant," B3 processing would update the retained candidates. In B2 processing, what appears to occur is a re-convergence toward a new single interpretation. Constructing a response while holding multiple candidates preserves room for those candidates to be updated later.

3-4. The Problem of Candidate Attenuation

What the examples in 3-1 through 3-3 have in common is not that the interpretation was wrong. The issue is that intent candidates may have become less likely to be referenced in processing beyond the next turn.

One premise worth confirming here is that human intent is not necessarily singular. Multiple intent purposes may exist simultaneously.

What B3 processing asks, then, is not "which candidate is correct." The question is whether the possibility that multiple intent purposes exist simultaneously can continue to be retained.

From this perspective, the problem of single-interpretation lock runs deeper than misinterpretation. It is not only that other candidates become less likely to be referenced once convergence toward the most probable candidate occurs. The possibility that multiple intent purposes exist simultaneously also becomes less likely to be referenced in subsequent processing.

This is referred to in this paper as the problem of candidate attenuation. When candidate attenuation occurs and the speaker indicates a different intent purpose in the next turn, what appears to happen is not an update of retained candidates but a re-convergence. And that re-convergence may itself be a convergence toward a new single interpretation.

Note that if candidate loss is not resolved, its effect may not be confined to a single exchange. This issue may manifest as particularly serious in contexts such as customer support. The possibility that candidate loss cascades into subsequent response costs is addressed in Chapter 7.

4. N=1 Observation Log — A Hypothesis That Deficits Create Detours

This chapter is a record of personal observation and does not aim at generalization.

4-1. N=1 Observation

It is possible that some characteristic existed in my process of understanding others' utterances.

This observation was not derived from theory. The observation that such a characteristic existed came first, and it was only later organized and conceptualized.

Note that there is no guarantee that introspection accurately reflects the underlying process. This is presented strictly as "something observed as such," and no external verification has been performed.

4-2. Hypothesis of Correspondence with B1

I observe that I had a tendency to interpret others' utterances literally.

This tendency may correspond to B1 as defined in this paper.

The author has been diagnosed with ASD, and a relationship with this tendency is conceivable. However, no causal relationship has been confirmed. What is observed at this point is the co-occurrence of an ASD diagnosis and this tendency; it cannot be said that ASD is the cause of the B1 tendency.

4-3. The Need to Bypass B2

I repeatedly received the comment "think about how others feel" from those around me.

However, I was unable to reproduce the understanding of others that those around me appeared to perform naturally. No matter how many times I received this feedback, the same failure recurred.

For this reason, I needed to attempt to understand others through a method different from the conventional one.

4-4. The B3 Formation Hypothesis

Looking back on this process, I observe that I came to move in the direction of considering what the other person wanted. It also appears that, in doing so, I was holding multiple candidates for communicative purpose.

In the course of trying to understand others through a different method, it is possible that I ended up moving toward placing weight on retaining candidates for communicative purpose. This paper refers to this formation process as the B3 formation hypothesis.

It is also possible that the axis of processing itself differed. Whereas B2's axis is "what is the optimal thing to return," what I moved toward was the axis of "what does the other person want."

However, this is an N=1 inference, and no comparative verification has been conducted.

4-5. Structural Comparison with LLMs

The following is a provisional model constructed from the observations and hypotheses in 4-1 through 4-4. It is presented not as a verified difference but as a hypothetical organization based on N=1 observation.

	LLM	Author (hypothesis)
B1	○	○
B2	Strong (optimized)	Possibility of insufficient functioning
B3	Weak (tendency to complete at B2)	Possibility of having been formed

5. Relationship with D2

5-1. What is D2?

In this paper, D2 refers to interpretive discrepancies that arise in dialogue and cognitive processing. D2 does not refer to a single misunderstanding or a single premise mismatch. Through repeated observation, it has become apparent that D2 may be a multi-layered cognitive misalignment that can occur simultaneously across multiple layers. The detailed structure of this model is left to a separate paper (the D2 paper).

This chapter examines the relationship between D2 and B3.

5-2. Relationship with Single-Interpretation Lock

The single-interpretation lock described in Chapter 2 may function as one of the pathways through which D2 arises.

When response candidates converge prematurely onto a single one and other candidates cease to be referenced, a gap between what the speaker intended to convey and the interpretation that was adopted can persist without being detected, even when such a gap exists.

This phenomenon can be considered one pathway that explains the occurrence of D2.

5-3. D2 May Not Be Misreading but Premature Hypothesis Finalization

D2 has conventionally often been explained as misreading or misinterpretation of an utterance. However, there is another way of viewing it.

What matters is not whether a candidate was deleted. What matters is that an interpretation was treated as a confirmed fact rather than being retained as a hypothesis.

As a result, other possibilities become difficult to reference in subsequent processing.

From this perspective, D2 can potentially be reinterpreted not as a problem of misreading, but as a problem of premature hypothesis finalization.

5-4. The Possibility That B3 Contributes to Reducing D2

As shown in 5-3, D2 can potentially be reframed not as a problem of misreading but as a problem of premature hypothesis finalization. From this view, the role B3 plays with respect to D2 can also be described differently.

B3 is the process of continuing to hold multiple hypotheses about communicative purpose. As long as a hypothesis is retained, an interpretation becomes less likely to be treated as the single confirmed fact.

For this reason, in contexts where B3-like processing is functioning, gaps at the level of communicative purpose may continue to be referenced as one among multiple candidates. This does not mean a function that accurately detects the gap. Rather than detection, what is maintained is a state in which the possibility of a gap continues to be referenced.

What matters here is that B3 is not a mechanism that prevents error. The candidates B3 retains may themselves be mistaken. What B3 may contribute is preventing an interpretation that could be mistaken from being finalized prematurely, leaving room for the candidate to be updated later.

B3, therefore, is positioned not as a means of resolving D2, but as something that may work to suppress premature hypothesis finalization, one of the conditions under which D2 arises.

6. Can B3-like Processing Be Implemented Through SCOPE × COS

6-1. The Role of SCOPE

The preceding chapters described B3 as the process of holding multiple hypotheses about communicative purpose. It is possible to generate candidates for communicative purpose without using the inferential structure called SCOPE. The role this paper assigns to SCOPE is not to handle the generation of candidates itself, but to provide a coordinate system for placing and continuing to observe the multiple intent hypotheses that B3 generates and holds.

SCOPE does not produce an answer. It is not a device for fixing where a given output originates as a single point. What SCOPE does is place multiple points-of-origin candidates together while preserving their relationships to one another. In practice, it can be used like a hypothesis board on which multiple hypotheses are arrayed and kept under continued observation.

SCOPE is not a diagnostic instrument. What SCOPE indicates is an inference about which layer a given output is likely to have arisen from; it is not a means of classifying or making definitive judgments about output.

In relation to B3, B3 is the process of generating and holding multiple intent hypotheses, while SCOPE can function as the coordinate system that places this set of hypotheses and sustains observation of them. What, then, is the significance of using SCOPE? This question is examined in 6-3.

6-2. The Role of COS

6-1 examined SCOPE. Whereas SCOPE is a coordinate system for placing and observing the intent hypotheses generated and held within B3, COS plays a different role.

COS is a framework that integrates multiple processing systems, including interpretation, decision-making, and communication. The process the B-series handles, from interpretation through response generation, is also part of the processing systems COS integrates. COS itself is not a function that carries out a single process; it is a framework for integrating these processing systems as a whole.

The precise hierarchical or dependency relationship between SCOPE and COS has not yet been fully worked out at the time of writing this paper. This paper treats SCOPE as an inferential coordinate system used in conjunction with COS.

In relation to B3, B3 is one pathway belonging to the B-series among the processing systems COS integrates. SCOPE placing and observing the intent hypotheses generated and held within B3 is positioned as one example of B3 and SCOPE functioning with distinct roles within the framework of COS.

6-3. The B3 External Implementation Hypothesis

The preceding sections examined B3 as the process of continuing to hold multiple hypotheses about communicative purpose, and SCOPE as the coordinate system for placing and continuing to observe those hypotheses. This section examines the possibility that combining these two could realize B3-like processing as an external structure.

B3-like processing may partially be possible within an LLM system alone. However, as described in Chapter 2, current LLMs appear to exhibit a tendency to converge prematurely onto the single most plausible candidate, even when multiple candidates are temporarily generated.

Using SCOPE in conjunction with this makes a different pathway possible with respect to this convergence. When a gap (D2) is observed in response to an utterance, multiple intent hypotheses regarding "why was this utterance generated" are first generated in a B3-like manner. This set of hypotheses is placed onto the coordinate system provided by SCOPE. SCOPE does not indicate which hypothesis is correct. What it enables is arraying multiple hypotheses together while preserving their relationships to one another, and continuing to observe them.

This observation does not end after a single instance. Even if a hypothesis comes to be strongly supported by new information, other hypotheses do not immediately disappear. The hypotheses continue to be updated while influencing one another. For example, when the plausibility of one hypothesis increases, the plausibility of a related hypothesis may be reconsidered accordingly; in this way, the set of candidates is treated not as a static list but as a network of observation in which elements act upon one another.

In the external implementation hypothesis addressed in this paper, combining these two is examined for its potential to make the retention and updating of multiple candidates easier to sustain.

What would happen if this approach were applied to an LLM has not been verified at this point. Possessing information about SCOPE and actually continuing to hold multiple candidates may be separate problems. This point is addressed again in Chapter 8.

6-3-1. Differences from Existing Research

Research dealing with multiple communicative intents also exists in the field of natural language processing. In such research, methods for detecting multiple intent candidates from a single utterance have been investigated.

There are both commonalities and differences between this existing research and the B3 hypothesis presented in this paper. The commonality is that both share the premise that multiple intents can exist behind a single utterance.

The difference lies in how the multiple candidates are treated. At least within the range the author has reviewed, candidate generation appears to be treated as an intermediate stage toward final classification or prediction.

By contrast, the B3 hypothesis examined in this paper does not aim at finalizing candidates. Even when new information comes to strongly support one candidate, the other candidates do not immediately disappear; they are carried forward into subsequent turns and continue to be held. This paper refers to this property not as candidate generation but as continued candidate retention.

Furthermore, as described in 6-3, the set of candidates assumed in this paper is not an isolated list. On the coordinate system provided by SCOPE, a change in the plausibility of one candidate may affect the plausibility of others, and the candidates are treated as a network of observation in which elements act upon one another. The set of candidates assumed in this paper appears, at least relative to the range of research the author has reviewed, to place greater emphasis on interaction among candidates.

In addition, the axis of communicative purpose addressed in this paper may itself differ from the intent categories assumed in existing research. At least within the range the author has reviewed, existing research appears to tend toward assuming a predefined, finite set of intent labels (such as booking, inquiry, or cancellation). The communicative-purpose candidates addressed in this paper, by contrast, are not selected from a fixed label set. Nor are they aimed at final confirmation; they are held as hypotheses that remain open to update.

6-4. Expected Behavioral Changes

This section organizes what behavioral changes could occur if the combination of SCOPE and B3 examined thus far were to actually function.

First, there is a change in interpretation. As described in 6-3, if candidates do not converge prematurely onto a single interpretation and continue to be held while preserving their relationships to one another, a state persists in which the communicative purpose behind the same utterance is not fixed to one. This does not mean that interpretive accuracy improves. What it means is that an interpretation becomes less likely to be treated as the sole confirmed one, and multiple possibilities continue to be referenced in parallel during that time.

Next is the change in response. As shown in Chapter 3, in B2 processing, a response is constructed based on the most plausible interpretation. In contrast, when candidate retention continuity is functioning, the response is not based solely on a single confirmed interpretation, but may be constructed in a way that takes into account the multiple candidates being retained. For example, in response to the utterance "Why did the data disappear?" discussed in 3-2, a response may become possible that takes into account both the candidate of seeking the cause and the candidate of expressing anxiety or distress. This is not a response that determines which candidate is correct, but a response premised on the possibility that multiple candidates may coexist. Retained candidates may also influence not only the content of a response but its strategy. Even when aiming for the same conclusion, how the dialogue subsequently unfolds may vary depending on which path is used to present it. When candidate retention continuity functions through B3 and SCOPE, a response may be constructed by taking into account not only its content, but also the possibility of continuing the dialogue and updating hypotheses afterward.

Finally, there is the reduction of D2. As described in 5-4, B3 is not a means of resolving D2, but may work to suppress premature hypothesis finalization, one of the conditions under which D2 arises. If B3-like processing combined with SCOPE can carry out this suppression more stably, gaps at the level of communicative purpose may become more likely to continue being referenced as one among multiple candidates, before being treated as a confirmed fact.

None of these changes have been verified at this point. What this paper presents is the direction of change expected when SCOPE and B3 are combined; confirming whether these changes actually occur requires the verification environment described in Chapter 8.

7. Possible Effects of B3-Type Processing

7-1. Basic Changes Brought About by B3

In the preceding chapters, this paper examined how B3 retains multiple hypotheses about utterance purpose, and its relationship to the inferential structure SCOPE. This chapter examines, in concrete terms, what kinds of changes may arise when this process functions.

As shown in Chapter 6, candidate retention continuity may have two aspects. One is a change in which the content of a response is constructed not on the basis of a single interpretation, but in a way that takes multiple candidates into account. The other is a change in response strategy: even when aiming for the same conclusion, how the dialogue subsequently unfolds may vary depending on which path is used to present it.

These two changes are not independent effects. Because candidates continue to be retained, an interpretation is prevented from being treated as a confirmed fact at an early stage. As a result, both the content of a response and the way it is presented come to be constructed while leaving multiple possibilities open.

The following sections examine the concrete forms this basic change may take across different contexts. First, in relation to D2 as discussed in Chapter 5, this paper considers how candidate retention continuity may relate to the early detection of misalignment in interpretation. Next, it considers how this detection may connect to a reduction in misalignment more broadly. It then turns to three areas of application — interpersonal support, education and coaching, and AI alignment — to examine what concrete forms this basic change may take in each.

It should be noted that both the results and the applications presented in this chapter indicate possible directions of change that may be expected if B3-type processing actually functions, and do not imply that such effects have been empirically demonstrated. What this chapter presents is an extension of the hypothesis; verification is addressed again in Chapter 8.

7-2. Early Reference to D2 and Reduction of Misalignment

As shown in Chapter 5, D2 can be reframed not as a problem of misreading, but as a problem of premature confirmation of a hypothesis. In situations where B3 functions, utterance purpose continues to be retained as multiple candidates, so before one interpretation is treated as a confirmed fact, possibilities that differ from it continue to be referenced in parallel.

This state does not imply a function that detects a discrepancy itself. As discussed in 5-4, what B3 may contribute is preventing the possibility of a discrepancy from being lost at an early stage, and preserving room for candidates to be updated later. In situations where this room is preserved, a discrepancy at the level of utterance purpose may become more likely to continue being referenced as the dialogue proceeds. This paper refers to this as early reference to D2.

The reduction of misalignment may arise through a separate mechanism. Whereas early reference to D2 concerns continuing to hold the possibility of a discrepancy before it is confirmed, the reduction of misalignment concerns the outcome of how that retained possibility is reflected in an actual response. As discussed in 7-1, candidate retention continuity may influence not only the content of a response but also its strategy. A response premised solely on one interpretation makes it difficult to continue the dialogue if that premise turns out to be wrong. A response that takes multiple candidates into account is more likely to leave room for the dialogue to continue without breaking down, even if a particular interpretation turns out to be incorrect.

Early reference to D2 and the reduction of misalignment can therefore be positioned as two distinct effects arising from the same mechanism of candidate retention continuity. The former concerns whether the possibility of a discrepancy continues to be referenced; the latter concerns the magnitude of the discrepancy's impact on the dialogue as a whole. The two may be linked, but one does not guarantee the other. Even if the possibility of D2 has been referenced from an early stage, misalignment may still occur if the subsequent response is constructed based on a single interpretation alone.

For this reason, this paper treats both within a single section, not as a single unified phenomenon, but as two distinct aspects of candidate retention continuity.

7-3. Application to Complaint Handling

The candidate retention continuity examined in the preceding chapters can take concrete form in interpersonal support contexts. This section examines this applied possibility through the example of complaint handling.

Consider the utterance "That's it, I'm done!" If the person handling the case interprets this as meaning the customer genuinely wants to end the interaction, the case is closed at that point. Afterward, however, dissatisfaction with the handling may surface later, in the form of an inquiry to head office or a public review. The interaction that was thought to have ended may not actually have ended.

Conversely, if the handler interprets this utterance as a strong emotional expression while the customer actually wants the explanation to continue, the explanation is continued. But this, too, can generate a new complaint: "I already said I was done, didn't I?"

Either response may appear reasonable on the surface. The issue is not the quality of the response itself. The issue is that the possibility of multiple utterance purposes coexisting behind a single utterance such as "That's it, I'm done!" was narrowed down to one at the time of the response.

In B2 processing, the response is constructed according to how the utterance is assigned meaning on the receiving end. What matters is not how the speaker actually felt, but how the receiver interpreted the utterance. A discrepancy may arise when the receiver's assigned meaning diverges significantly from the speaker's utterance-purpose candidates.

Behind this utterance, multiple utterance purposes may exist simultaneously: genuinely wanting to end the interaction; having chosen a strong emotional expression while actually wanting the interaction to continue; having lost trust in the person handling the case; or seeking a different solution altogether. These are only some of the candidates that may arise from the same utterance. These candidates are not necessarily independent of one another. Distrust toward the handler may manifest as a desire to end the interaction, or it may manifest as a demand for a different solution.

In situations where B3 processing functions, none of these candidates is prematurely fixed as the correct one; the response is constructed while holding multiple possibilities. What such a response aims for is not determining which candidate is correct, but finding a form that allows the dialogue to continue regardless of which candidate turns out to be true.

For example, a response such as "Would you like me to transfer you to a different representative?" can be considered. This response does not exclude the candidate of genuinely wanting to end the interaction. For the candidate of wanting the interaction to continue despite the emotional phrasing, it offers an opportunity to reset the situation. For the candidate of having lost trust in the handler, it responds directly to that distrust. In other words, this response is constructed without presupposing a single candidate, in a way that allows the dialogue to proceed regardless of which candidate turns out to be correct.

The reaction to this response functions as new information. A reply such as "That's not it" suggests an utterance purpose different from the initial candidates. A reply such as "Let me speak to your supervisor" strongly supports the candidate of distrust toward the individual handler. Replies such as "It won't matter who handles it" or "I'm just going to leave" strongly support the candidate of genuinely wanting to end the interaction, or distrust toward the handling process as a whole. In each case, the candidates are not finalized but updated by new information.

In this way, response generation under B3 processing may take the form of a repeating cycle: utterance, candidate retention, response, and candidate updating through the next utterance. As discussed in 7-1, this concerns not only the content of the response but also the path by which it is presented. The aim of the response is not to resolve the issue immediately, but to obtain, while holding multiple candidates, the information needed to discern which candidate will be supported next.

It should be noted that the discrepancy behind such utterances may not be limited to the level of utterance purpose, and may have a more multi-layered structure. Even utterances that appear on the surface as the same kind of "dissatisfaction" may involve multiple distinct layers in their occurrence; this point is examined in a separate paper (the D2 paper).

7-4. Application to Education and Coaching

The case of complaint handling examined in 7-3 was an example in which candidates were updated within a single dialogue. In educational and coaching contexts, because dialogue with the same person continues over time, the range in which candidate retention continuity operates may extend across a much longer time frame.

Consider an utterance frequently encountered in educational and coaching contexts: "I understand." If the instructor interprets this as meaning the learner has genuinely understood, the discussion moves on to the next topic. However, in a subsequent evaluation or practical exercise, it may become apparent for the first time that the learner had not actually understood.

It should be noted here that this discrepancy does not necessarily arise solely from a misjudgment on the instructor's part. In educational and instructional settings, it is often observed that the speaker themselves mistakenly believes they have understood. The speaker is not lying; they genuinely feel, at that moment, that they have understood. Yet at a later stage of practice or application, it may become apparent, even to the speaker themselves, that their understanding had been insufficient. In this case, the discrepancy exists not only as an error in the instructor's assigned meaning, but also as an error in the speaker's own self-perception.

In either case, the exchange may appear, at the time, to have been completed without issue. Yet the discrepancy surfaces only later, during evaluation or practice.

Behind the utterance "I understand," there may exist a discrepancy in the speaker's own perception of their understanding: genuinely having understood, versus mistakenly believing one has understood despite insufficient comprehension. At the same time, there may also exist utterance purposes such as not wanting to admit a lack of understanding, being concerned that one's evaluation will suffer, or wanting to move the discussion along quickly.

In situations where B3 processing functions, none of these candidates is prematurely fixed as the correct one; the subsequent response is constructed while holding multiple possibilities. What such a response aims for is not determining which candidate is correct, but finding a form that allows the next exchange to help distinguish between them.

For example, a response such as "Could you try explaining what we just covered in your own words?" can be considered. For the candidate of genuine understanding, this functions as a confirmation of that understanding. For the candidate of mistakenly believing one has understood despite insufficient comprehension, this may function as an opportunity for the learner to reassess their own understanding. For the candidate of being concerned about evaluation, the fact that this is presented as a check rather than a judgment may leave room for an easier response. For the candidate of wanting to move on quickly, the response proceeds in a direction different from that intent, but the subsequent reaction increases the likelihood that this candidate will be supported.

Another aspect of candidate retention continuity in educational and coaching contexts is the length of the time frame involved. In complaint handling, candidates tend to be updated within a relatively short period, within a single dialogue. In education and coaching, candidates may remain unresolved within a single session and instead be gradually narrowed down over the course of multiple sessions. The candidate of "having understood" may be supported in one session, while the candidate of "having understood insufficiently" may be supported in another, such that the support relationships among candidates continue to be updated across multiple dialogues.

For this reason, if B3 processing functions in educational and coaching contexts, it may manifest not only as candidate updating within a single dialogue, but also as long-term candidate retention sustained across multiple dialogues.

7-5. Implications for AI Alignment

As discussed in 7-1, what a single-interpretation lock fixes in place is not interpretation alone. The path of the response generated on the basis of that interpretation is also fixed simultaneously. This may occur in AI systems in the same way.

Consider a situation in which a user strongly asserts a claim that contains logical leaps, or a claim on which ethical disagreement is likely to arise. If the AI converges early on a single interpretation of that claim and returns a correction or denial based on that interpretation, the dialogue is likely to move in a direction that closes it off, even if the response is logically or ethically sound. Why the user arrived at that claim, and what experiences or premises underlie it, are left without being referenced. This is structurally the same as the response to "That's it, I'm done!" examined in 7-3, where a single interpretation was treated as the basis for the response, and the response to "I understand" in 7-4, where the utterance was taken at face value.

When B3 processing functions, the response is not constructed on the basis of a single confirmed judgment, but rather while holding multiple utterance-purpose candidates, in a form that allows the next piece of information to be obtained. For example, responses that ask what led the user to that claim, or what kind of situation they have in mind, can be considered. These responses do not immediately confirm which candidate is correct. They are responses designed to obtain the user's reply itself as new information for narrowing down the candidates.

This difference may provide a perspective for reconsidering AI alignment not only at the abstract level of safety, but at the more concrete level of dialogue design and Human-AI Interaction. As consistently seen across 7-3 and 7-4, what B3-type processing aims for is not arriving at the correct interpretation, but leaving in place, within the response itself, a path that allows the dialogue to continue regardless of which candidate turns out to be correct. Whereas a single-interpretation lock narrows the path of a response, B3-type processing works in the direction of preserving the options along that path.

This is not a mechanism that immediately resolves misalignment. As discussed in 5-4, B3 is not a mechanism for preventing errors, and the possibility always remains that the retained candidates themselves are incorrect. However, in that it prevents the dialogue path from being fixed early to a single option and preserves room to continue referencing the user's theoretical structure and premises, B3-type processing may be relevant to the continuity of the dialogue itself.

It should be noted that actual cognitive misalignment may have a more multi-layered structure that is not limited to the level of utterance purpose. This point is examined in a separate paper (the D2 paper).

It should also be noted that how the response generated by B3 is conveyed to the other person falls within the domain of the C Series (Communication Strategy Process) in COS. As this paper focuses on B3, the C Series will be examined in a separate paper.

8. Habitat Requirements

The hypotheses presented in this paper cannot be verified through individual observation alone. Verifying the B3 hypothesis requires an environment (habitat) in which continuous observation and comparative experimentation are possible.

8-1. Why a Habitat Is Necessary

The B3 hypothesis presented throughout this paper remains unverified at this stage. What has been observed is the possibility that a process exists in which multiple hypotheses about utterance purpose continue to be retained. Whether that process actually functions, under what conditions it functions, and to what degree of effect, have not yet been confirmed.

Verifying this hypothesis requires more than isolated observations. In order to distinguish whether B3-type processing is functioning as candidate retention continuity, or whether similar results are arising through a different process, an environment is needed in which what is occurring during the course of a dialogue can be recorded externally and cross-referenced afterward. In addition, in order to compare cases in which B3-type processing functions with cases in which it does not, an environment is needed in which conditions can be set for different processes to operate on the same utterance.

Furthermore, in order to confirm whether the applied possibilities examined in Chapter 7 — complaint handling, education and coaching, and AI alignment — actually function in practice, continuous observation and the accumulation of failure cases in each of these domains will be necessary.

The following sections set out the requirements for a habitat needed to verify the B3 hypothesis.

8-2. Utterance-Purpose Cross-Reference Environment

One of the most important requirements for verifying the B3 hypothesis is an environment in which utterance purposes can be cross-referenced.

The utterance purposes that B3 deals with cannot be directly observed from the outside. This is both a foundational premise of the B3 hypothesis and the greatest factor making verification difficult.

In order to partially resolve this difficulty, an environment is needed that allows for cross-referencing through post-dialogue reports by the speaker and through long-term observation. For example, if a system exists in which, after a dialogue ends, the speaker can be asked "What were you really trying to convey with that utterance?" or "What were you seeking at that point?", it becomes possible to cross-reference the candidate set that B3 had retained against the utterance purpose reported by the speaker.

However, this cross-referencing has its limits. There is a possibility that the speaker themselves does not accurately grasp their own utterance purpose. As examined in 7-4, cases in which a person's self-perception diverges from their actual state have been observed in educational and coaching contexts as well. The possibility cannot be excluded that a report of utterance purpose is a post-hoc reconstruction. Reports by the speaker need to be treated not as the correct answer, but as a reference point for cross-referencing.

Even so, an utterance-purpose cross-reference environment is indispensable. Even if complete cross-referencing is impossible, confirming whether a systematic correspondence exists between the candidate set and the report constitutes the most critical observation point in verifying the B3 hypothesis.

8-3. Failure Log Collection Environment

In verifying the B3 hypothesis, failure cases are at least as important as success cases, if not more so.

The candidate attenuation problem and single-interpretation lock examined in this paper are difficult to observe in situations where dialogue "goes well." Discrepancies surface in situations where dialogue breaks down because candidates were eliminated at an early stage, or where it later becomes apparent that a different utterance purpose had been correct. In other words, cases in which B3-type processing did not function constitute the most important objects of observation for verifying the hypothesis.

Of particular importance are logs of dialogues in which D2 was observed. As discussed in Chapter 5, premature confirmation of a hypothesis may be one factor contributing to D2. By tracing back through dialogues in which D2 was observed, it may become possible to confirm at what point and which candidates ceased to be referenced. The accumulation of such D2 logs will serve as material for identifying the conditions under which candidate retention continuity did not function.

However, collecting failure logs also has its limits. If the participants in a dialogue do not recognize the interaction as a failure, it may not be recorded as a log. In addition, distinguishing whether the cause of a failure was the absence of B3-type processing or some other factor will require use in combination with other requirements.

The accumulation of failure logs forms the foundation for clarifying the conditions under which the B3 hypothesis functions and those under which it does not. Success cases alone cannot explain when and why B3-type processing fails to function.

8-4. Hypothesis Comparison Environment

In order to confirm whether candidate retention continuity is actually occurring in verifying the B3 hypothesis, an environment is needed in which the dialogue process can be compared and cross-referenced from the outside.

Of particular importance is an environment that can distinguish between candidate retention continuity and post-hoc generation. Asking "What were the initial candidates?" after a dialogue has ended is insufficient to determine whether candidates were actually retained throughout, or whether they were newly generated at that point. In order to confirm whether candidate retention continuity truly occurred, an observation environment is needed in which candidates are recorded externally on a turn-by-turn basis during the course of the dialogue, and can be cross-referenced afterward.

In addition, in order to compare how B1, B2, and B3 processing each generate different responses to the same utterance, an environment is needed in which the conditions of processing can be controlled. In the course of natural dialogue, it is difficult to determine from the outside which processing is taking place. Only with an environment in which comparative experimentation is possible can it be confirmed whether B3-type processing produces results that differ from other types of processing.

It should also be noted that by combining turn-by-turn external records with other requirements, it may become possible to observe the correspondence between candidate retention continuity and speaker reports.

8-5. Long-Term Observation Log Environment

In verifying the B3 hypothesis, there are phenomena that cannot be seen through isolated observations of individual dialogues. The effects of candidate retention continuity may be observed not only within a single dialogue, but across multiple dialogues. As examined in 7-4, in educational and coaching contexts, the support relationships among candidates may continue to be updated across multiple dialogues. Observing this kind of phenomenon requires a continuous observation environment sustained over a long period of time.

There is a second reason why long-term observation is necessary. The effects of B3-type processing failing to function may not surface immediately. As with the candidate attenuation problem discussed in 3-4, discrepancies may accumulate over the course of repeated dialogues and surface later in a different form. Isolated observations cannot capture this accumulation process.

What is required of a long-term observation log environment is a system in which dialogue records are retained in chronological order and can be traced back to a specific point in time for cross-referencing afterward. When it appears that candidates ceased to be referenced at a particular turn, it becomes important to be able to trace back and observe the relationship with dialogues that preceded that point.

However, long-term observation requires ethical consideration. The collection and retention of dialogue logs is contingent on the consent of the dialogue participants and on a data management policy. These constitute the conditions for the long-term observation environment itself to be established. This point needs to be addressed in advance in the design of the habitat.

8-6. High-D2 Observers

In verifying the B3 hypothesis, not only the observation environment but also the quality of the observer may be an important requirement.

D2 may be sensed prior to classification. An unresolved sense of discrepancy — a feeling that "something is off" — may fire first, with classification and verbalization following afterward. Whether this sensing is triggered varies by observer. Even in the same dialogue, an observer with a high sensitivity threshold may be able to detect discrepancies at an early stage, while an observer with a lower threshold may not trigger sensing at all until the discrepancy has become substantially apparent. This point will be examined in detail in a separate paper.

In verifying the B3 hypothesis, the presence of observers capable of sensing D2 with high sensitivity may be useful. In order to identify the point at which candidates cease to be referenced, it may be required that observers capable of sensing where a discrepancy arose within a dialogue are able to record and verbalize that discrepancy. Whether the same observation is possible through mechanical log collection alone is not clear at this stage.

However, this kind of observer requirement also has its limits. The observer's own premises and state may influence the observation results. Estimation through SCOPE may carry observer dependency. For this reason, observation by multiple observers from different angles may have higher reliability than observation by a single observer.

The observations in this paper are based on N=1 observation by a single observer. The validity of the concept of a high-D2 observer itself will also require future verification. Verification by multiple observers is related to the researcher network discussed in 8-8.

8-7. Implementation Engineers

In verifying the B3 hypothesis, the presence of implementation engineers may be necessary in order to build the theoretical framework into an actually functioning system.

The B3 external implementation hypothesis examined in Chapter 6 indicated the possibility that B3-type processing could be realized not only within the internal structure of an LLM, but as an external structure. However, in order to confirm whether this possibility actually functions, an environment is needed in which a prototype can be built and its operation observed. With theoretical description alone, it may be difficult to confirm whether candidate retention continuity is actually occurring within a system.

For this purpose, the ability to understand the concept of B3-type processing and translate it into a system in an observable form will be necessary. In particular, the turn-by-turn external recording discussed in 8-4, and the mechanisms for distinguishing between candidate retention continuity and post-hoc generation, require implementation-level design.

However, the role of implementation engineers may not be limited to prototype construction. In the course of implementation, problems that had not been anticipated at the theoretical level may be discovered. In this sense, implementation engineers may function not only as builders of the verification environment, but also as observers involved in the refinement of the hypothesis.

8-8. Researcher Network

In verifying the B3 hypothesis, there are aspects that cannot be fully captured by a single observer or a single area of expertise. By bringing together researchers with different specializations and perspectives, multi-faceted verification of the hypothesis may become possible.

The B3 hypothesis examined in this paper involves problems that span multiple domains, including cognitive science, linguistics, natural language processing, HCI, and dialogue design. For example, the question of how candidate retention continuity is realized cognitively is relevant to the domain of cognitive science. The possibility of implementation in LLMs is relevant to the domain of natural language processing. The effects on dialogue continuity and UX are relevant to the domains of HCI and dialogue design. It is not easy for a single researcher to cover all of these.

In addition, as discussed in 8-6, there is the problem of observer dependency. In order to exclude the possibility that the observation results of a single observer are influenced by that observer's own premises and state, observations that can be mutually cross-referenced by multiple researchers with different premises and specializations may be necessary.

Furthermore, critical examination of the hypotheses presented in this paper is also one of the important functions of a researcher network. The presence of researchers who can identify weaknesses and oversights in the hypothesis may play an indispensable role in refining it.

This paper began as a hypothesis arising from N=1 observation. For this hypothesis to develop into a verifiable theory, not only a researcher network capable of observation and critical examination from different perspectives, but also the construction of a habitat that integrates the requirements set out in 8-2 through 8-7, may be necessary.

8-9. Funding

The requirements set out in 8-2 through 8-8 all presuppose the construction and continuous operation of a verification environment. For this reason, the foundations needed to actually maintain these requirements also need to be considered. A funding base to support continuous operation may be necessary.

In particular, long-term observation (8-5) and the researcher network (8-8) may be difficult to address through one-off projects. Verifying the B3 hypothesis may require an environment in which observation spanning multiple dialogues and the continuous involvement of researchers from different specialized domains are both possible. These are not well suited to verification that concludes within a short period of time.

At this stage, a funding base capable of sustaining the continuous operation of the habitat envisioned in this paper has not been established. This current situation is addressed in 8-10.

8-10. Current Status

The B3 hypothesis presented in this paper remains unverified at this stage.

What has been observed is the possibility that a process exists in which multiple hypotheses about utterance purpose continue to be retained. Whether that process actually functions, under what conditions it functions, and to what degree of effect, have not yet been confirmed. This paper is a record of observation and reasoning, not a verified theory.

The habitat requirements set out in 8-2 through 8-9 represent an attempt to organize the conditions necessary for this hypothesis to develop into a verifiable theory. An utterance-purpose cross-reference environment, a failure log collection environment, a hypothesis comparison environment, a long-term observation log environment, high-D2 observers, implementation engineers, a researcher network, and a funding base — none of these are currently in place.

However, this current situation does not mean that the B3 hypothesis is unobservable. Even a hypothesis that began from N=1 observation, if it points to an observable phenomenon and has been described in a form that can be verified, holds the possibility of being verified under conditions in which a habitat becomes established.

What this paper presents is not verified fact, but a hypothesis awaiting verification. The observations and reasoning are recorded here, in preparation for the day when the verification environment is in place.